

Scalable Static Website with Azure Storage + Vercel + GitHub Actions

Objective

The main goal of this project is to host a simple, scalable, and fast-loading static website using Azure storage's free tier. To make the site available globally and secure with HTTPS, Vercel's free services are used. The deployment is fully automated using GitHub Actions, so every time I update the website's content in the GitHub repository, the changes are pushed to the live website without manual effort.

Tools Used

- Azure Storage – To store the website files like HTML and CSS.
- Vercel (Free) – To provide a global content delivery network (CDN), manage DNS, and enable HTTPS.
- GitHub Actions – To automate the process of updating the website whenever I push new changes to GitHub.
- HTML/CSS – For designing the structure and style of the website.

Steps to Implement the Project

1. Creating the Website

I designed a basic website with two pages using HTML and CSS. The pages have a clean layout and include headings, paragraphs, and navigation links. Once the website structure was ready, I pushed it to a GitHub repository.

2. Setting up Azure Storage

I created a new Azure Storage with a unique name and enabled public access to serve static content. I configured the bucket for static website hosting, specifying index.html as the default page. Permissions were adjusted using a bucket policy so that the files can be accessed globally.

3. Connecting GitHub Actions

I created a `.github/workflows/deploy.yml` file to define the automated workflow. This workflow is triggered on every commit pushed to the main branch. It installs the Azure CLI and syncs the website files to the Azure Storage using the `Azure Storage sync` command. Azure Storage credentials were securely stored in GitHub Secrets to authenticate the deployment process.

4. Integrating with Vercel

I added my custom domain to Vercel and updated the DNS settings to point to the Azure Storage website endpoint. I enabled “Always Use HTTPS” in Vercel settings to ensure secure browsing.

Cache rules were applied to improve performance and reduce unnecessary requests to the Azure Storage. I also configured versioning by appending timestamps in the URL to prevent caching issues during updates.

Results

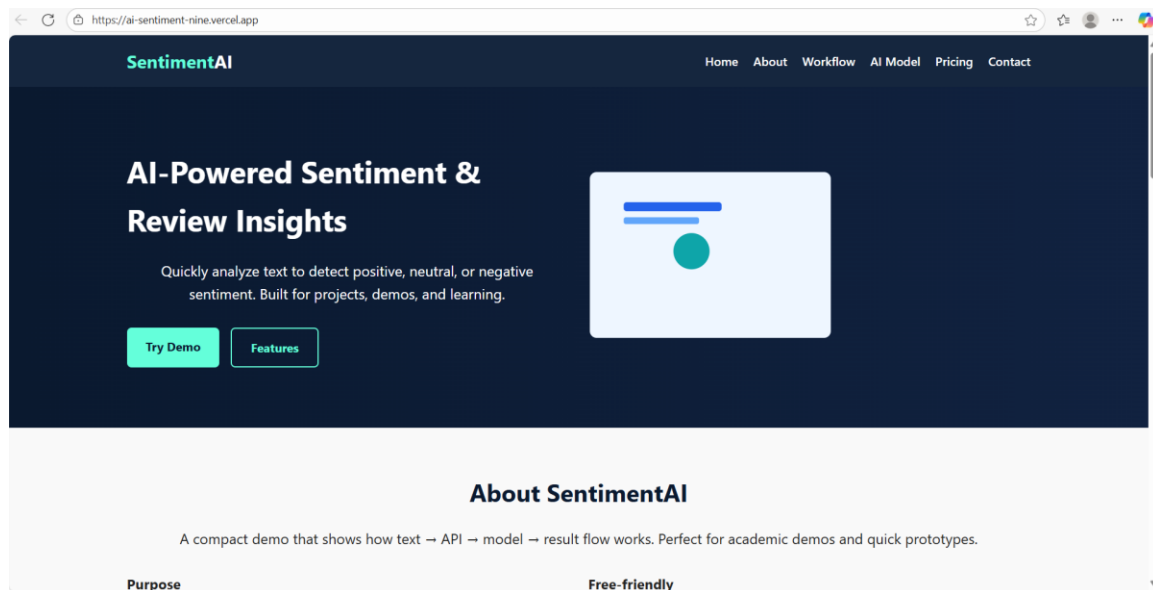
The final setup is working smoothly:

- Whenever I push changes to the GitHub repository, the website is automatically updated within seconds.
- The site is accessible globally via HTTPS.
- The content is delivered fast through Vercel's CDN.
- Versioning and cache rules ensure that users always see the latest content.

Live Website Link

<https://ai-sentiment-nine.vercel.app/>

Screenshot



Conclusion

This project taught me how to build a fully automated, secure, and scalable static website using popular tools. With **Azure Storage**, Vercel, and GitHub Actions, I was able to create a deployment pipeline that is easy to manage, fast, and globally available. The setup can be extended for more complex projects and provides a strong foundation for web development workflows.